

WEEK 1 · DAY 1

What Is Cloud Computing?

BaseStack Academy · AWS Cloud Accelerator

Cloud Foundations

Today's Agenda

What Is Cloud Computing?

1

Overview — Why This Matters

FOUNDATION

2

Key Concepts — Definitions & Vocabulary

CORE KNOWLEDGE

3

Deep Dive — Key Concepts Extended

DEEPER LEARNING

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How It Works — Step by Step

MECHANICS

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Technical Deep Dive

ADVANCED DETAIL

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Architecture & Visual Model

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Real-World Nigerian Scenario

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Common Mistakes & Exam Traps

EXAM PREP

WEEK 1 · DAY 1

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Overview — Why This Matters

Cloud computing has fundamentally transformed how businesses operate globally. Instead of purchasing expensive physical servers and maintaining data centres, organisations can now rent computing resources on-demand from providers like AWS. This shift has democratised access to enterprise-grade technology — a startup in Lagos can now use the same infrastructure as a Fortune 500 company in New York.

For you as an aspiring cloud professional, understanding cloud computing is not just a technical necessity — it is a career lifeline. Africa's tech sector is growing at an unprecedented rate, and cloud skills are consistently ranked among the most in-demand capabilities.

WHY NOW?

Cloud skills are the #1 most in-demand tech capability globally. Understanding this topic is essential for both the SAA-C03 exam and real cloud engineering work.



Beginner Note

Do not worry if this feels new — every expert was once a complete beginner. Take notes and ask questions freely.

AWS Advantages

-  1 Trade capital expense (CapEx) for variable expense (OpEx).
-  2 Benefit from massive economies of scale.
-  3 Stop guessing capacity.
-  4 Increase speed and agility.
-  5 Stop spending money running and maintaining data centres.
-  6 Go global in minutes.



Key Concepts — Know These Definitions

 Study tip: Say each definition aloud — spoken repetition builds memory 3× faster than reading alone.

On-Demand Self-Service

Provision resources instantly with no human interaction from the provider

Broad Network Access

Access services from any device over standard internet connections

Resource Pooling

Multiple customers share the same physical infrastructure securely via multi-tenancy

Rapid Elasticity

Scale resources up or down within minutes — or automatically — based on demand

Measured Service

Pay only for what you consume, like a utility bill — electricity or water





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INSTANT EXPANSION
+300 GUESTS? NO PROBLEM

AS E DEY HOT!
BURNÀ BOY

Key Concepts (Continued)

Concepts Summary

All key concepts for this lesson have been covered. Use this slide to review and quiz yourself.

✓ On-Demand Self-Service

✓ Broad Network Access

✓ Resource Pooling

✓ Rapid Elasticity

✓ Measured Service



How It Works — Step by Step

1

Cloud computing works through massive, highly secure data centres maintained by providers like Amazon Web Services. These facilities house thousands of physical servers connected by high-speed networks. Using virtualization technology, each physical server is divided into many virtual machines that can be rented individually. When you create an EC2 instance on AWS, you are renting a slice of a physical server located in an AWS data Centre.

2

The three primary service models are: Infrastructure as a Service (IaaS) — you rent raw computing resources and manage everything about the hardware; Platform as a Service (PaaS) — you get a managed environment to deploy applications without managing servers; and Software as a Service (SaaS) — you use finished applications like Gmail, Dropbox, or Google drive with zero infrastructure.

3

The four deployment models are: Public Cloud (shared, managed by AWS), Private Cloud (dedicated to one organisation), Hybrid Cloud (mix of on-premises and cloud), and Multi-Cloud (using multiple cloud providers simultaneously).



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PDU-B18





How It Works — Step by Step

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Understanding the Fundamentals

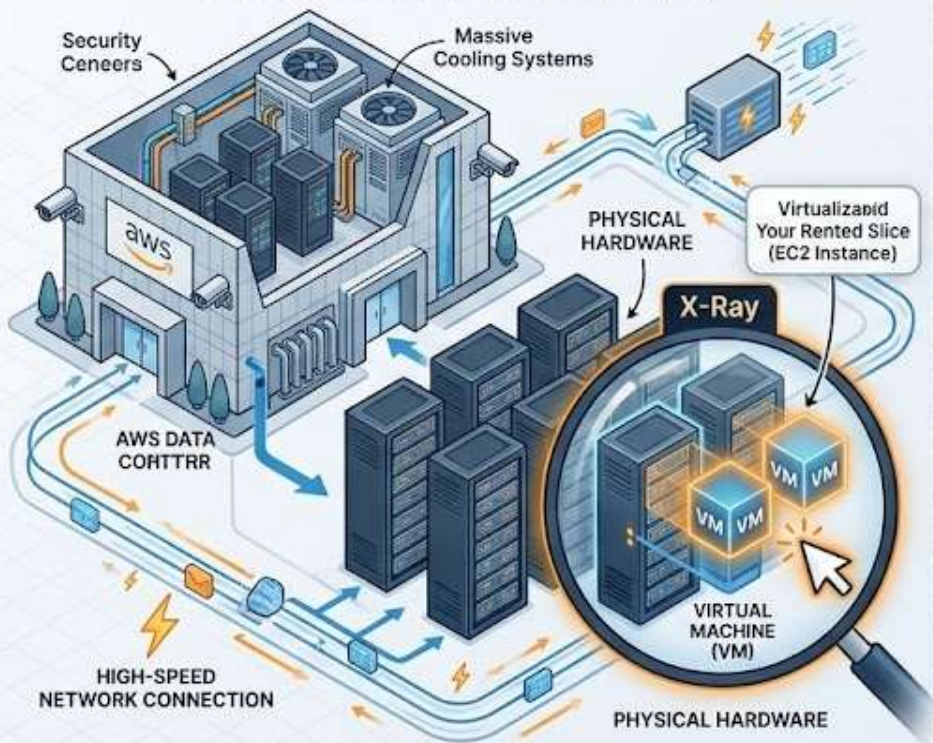
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TECHNICAL DEEP DIVE: UNDERSTANDING THE FUNDAMENTALS

The Physical Cloud

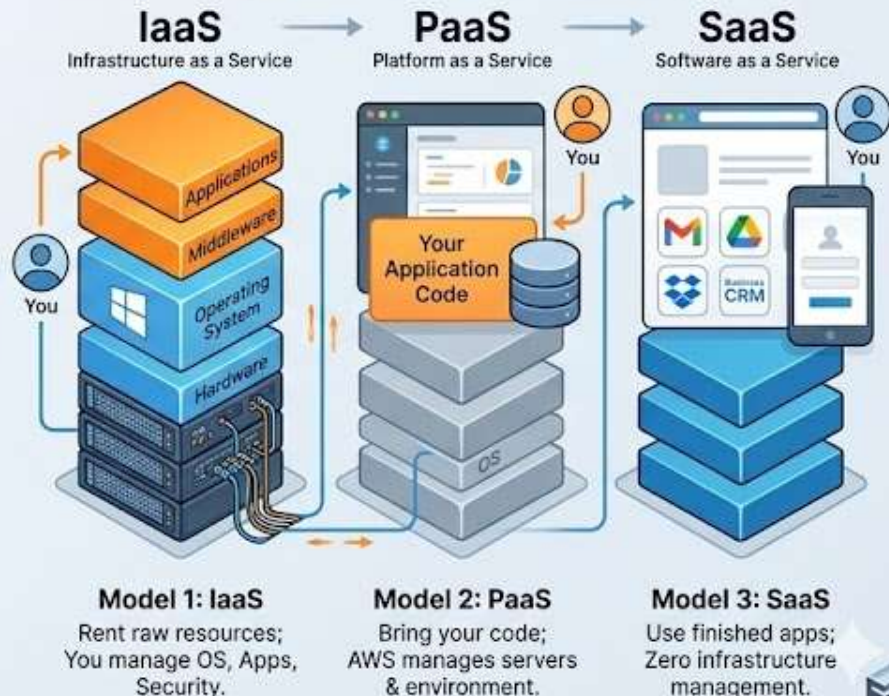
DATA CENTERS & VIRTUALIZATION FLOW



The 3 Service Models

SHARED RESPONSIBILITY STACK (Decreasing User Management)

Managed by You (Orange) Managed by AWS (Blue)





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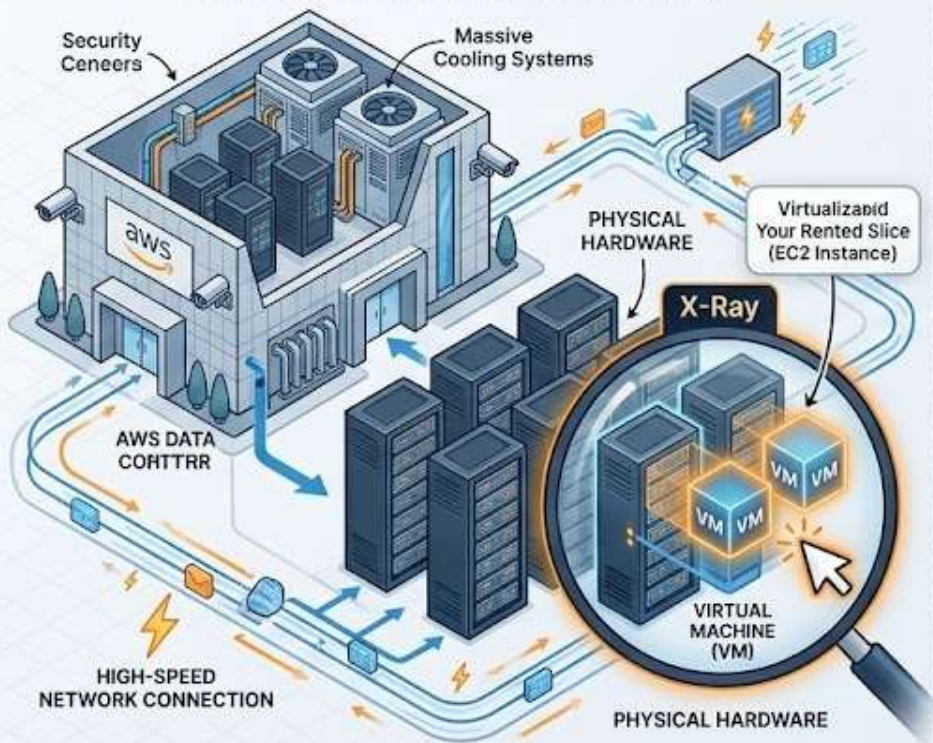
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TECHNICAL DEEP DIVE: UNDERSTANDING THE FUNDAMENTALS

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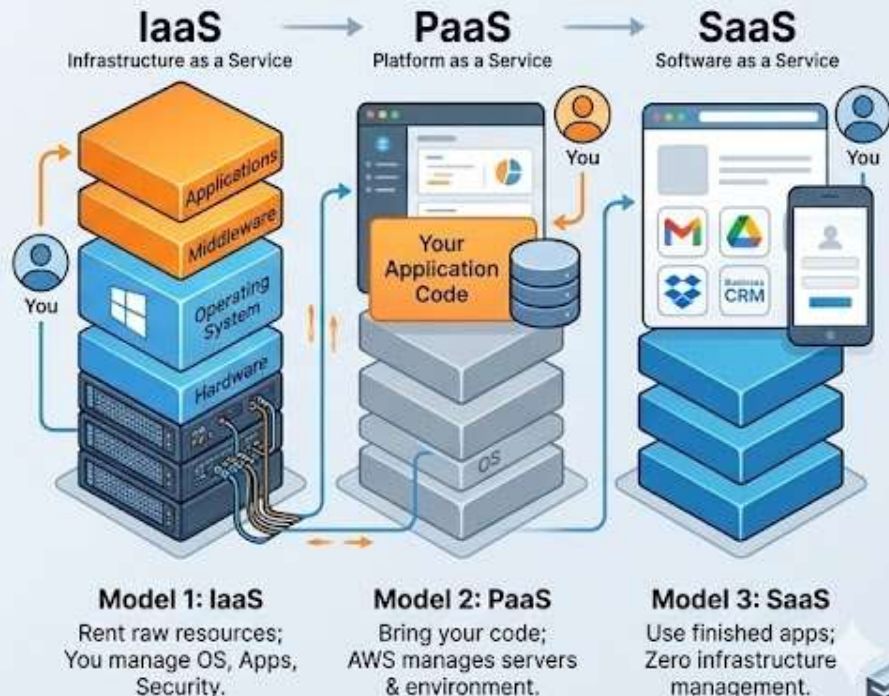
DATA CENTERS & VIRTUALIZATION FLOW



The 3 Service Models

SHARED RESPONSIBILITY STACK (Decreasing User Management)

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Architecture & Visual Model

Core Components

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How They Connect

Rapid Elasticity

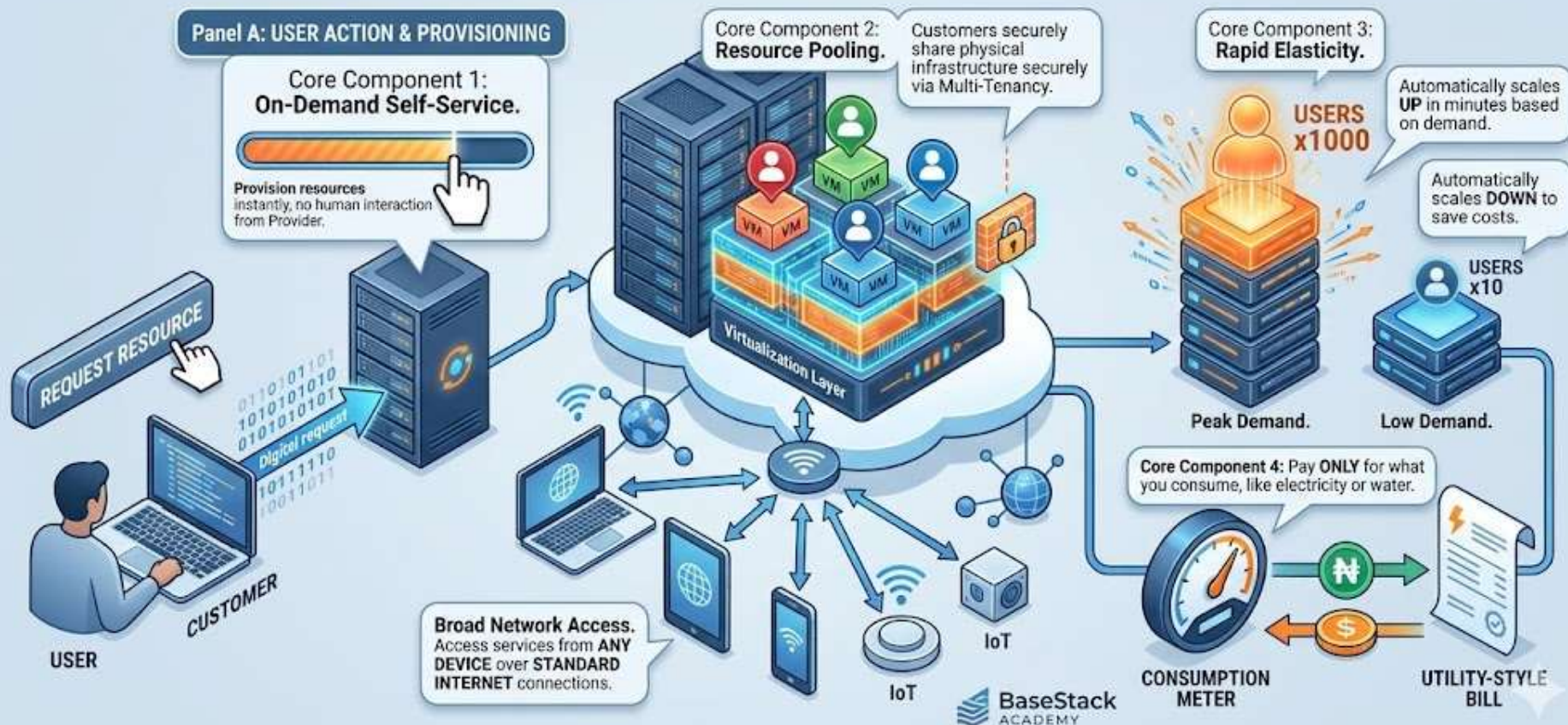
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Measured Service

Pay only for what you consume, like a utility bill — electricity or water

CLOUD ARCHITECTURE VISUAL MODEL: HOW COMPONENTS CONNECT

TECHNICAL DEEP DIVE: CORE CHARACTERISTICS IN MOTION



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Real-World Nigerian Scenario

Consider Chidi, who runs a Nigerian fintech startup processing mobile payments. Before the cloud, he needed ₦15 million to buy servers, hire a systems administrator, and rent data centre space — and he still had to guess how much capacity to buy. Using AWS, he spends only what he needs each month. When his app goes viral during a Christmas promotion, AWS automatically handles the traffic surge. On quiet Tuesday mornings, costs drop to near zero. He never pays for idle hardware. This is cloud computing

Discussion Question

Think about a Nigerian business you know. How would this AWS service change how they operate? What problem would it solve?





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Discussion Question

Think about a Nigerian business you know. How would this AWS service change how they operate? What problem would it solve?





Common Mistakes & Exam Traps



These are the most common exam errors — every one costs marks. Memorise them.



Cloud computing has FIVE essential characteristics per the NIST definition — not three or four. Memorise: On-Demand, Broad Network Access, Resource Pooling, Rapid Elasticity, Measured Service.



Know the distinction between IaaS (you manage OS), PaaS (provider manages OS + runtime), and SaaS (provider manages everything). This distinction is tested repeatedly.



On-Demand pricing has NO upfront commitment. Reserved Instances and Savings Plans require commitment in exchange for discount.



Question

A Solutions Architect at a Lagos fintech company is designing the AWS infrastructure for a new payment platform. They need to implement best practices related to What Is Cloud Computing?. Which of the following statements is CORRECT?

- A** Provision resources instantly with no human interaction from the provider
- B** Cloud computing has FIVE essential characteristics per the NIST definition — not three or four....
- C** Know the distinction between IaaS (you manage OS), PaaS (provider manages OS + runtime), and...
- D** On-Demand pricing has NO upfront commitment. Reserved Instances and Savings Plans require...



SAA-C03 Practice Question

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Question

A Solutions Architect at a Lagos fintech company is designing the AWS infrastructure for a new payment platform. They need to implement best practices related to What Is Cloud Computing?. Which of the following statements is CORRECT?

A

Provision resources instantly with no human interaction from the provider

✓ CORRECT

B

Cloud computing has FIVE essential characteristics per the NIST definition — not three or four....

C

Know the distinction between IaaS (you manage OS), PaaS (provider manages OS + runtime), and...

D

On-Demand pricing has NO upfront commitment. Reserved Instances and Savings Plans require...



Key Takeaways — What You Learned Today

What Is Cloud Computing?

Concepts Covered

✓ On-Demand Self-Service

✓ Broad Network Access

✓ Resource Pooling

✓ Rapid Elasticity

✓ Measured Service

Before Next Class

1. Review the concepts above without notes

2. Complete the hands-on console lab

3. Post your progress on LinkedIn / X

4. Review the exam traps from the previous slide

5. Check the Further Resources list



Further Resources & Reading

Go deeper with these official AWS resources:

<https://aws.amazon.com/what-is-cloud-computing/>

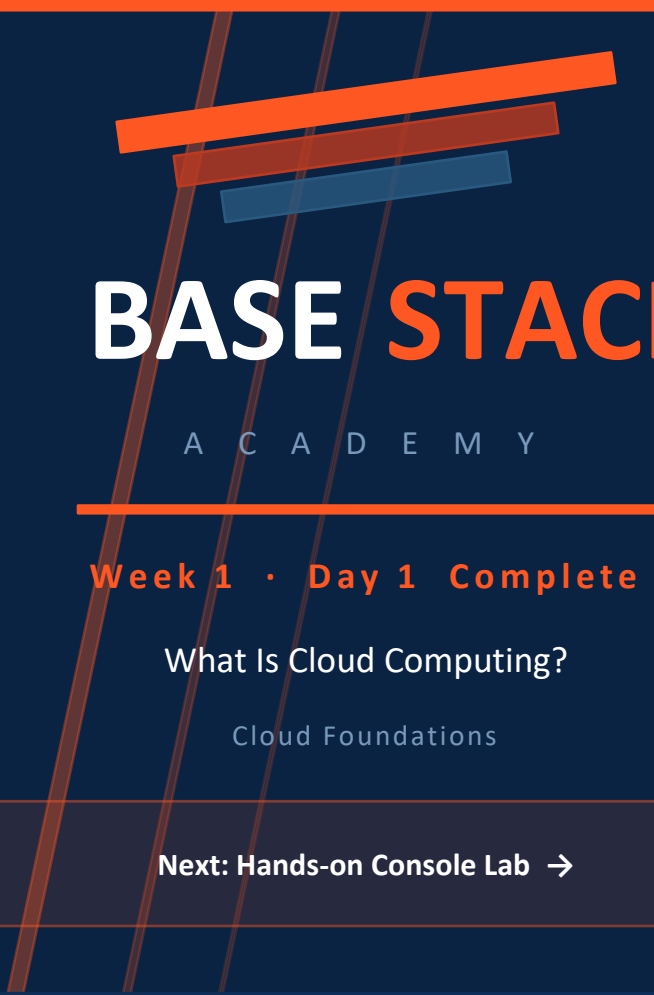
<https://docs.aws.amazon.com/whitepapers/latest/aws-overview/introduction.html>

https://www.youtube.com/watch?v=M988_fsOSWo

<https://aws.amazon.com/types-of-cloud-computing/>



Homework: Read at least ONE link above before the next class. Post what you learned on the WhatsApp group.



BASE STACK

A C A D E M Y

Week 1 · Day 1 Complete

What Is Cloud Computing?

Cloud Foundations

Next: Hands-on Console Lab →